PROBLEM FS9

Hinged Gate Opening Height

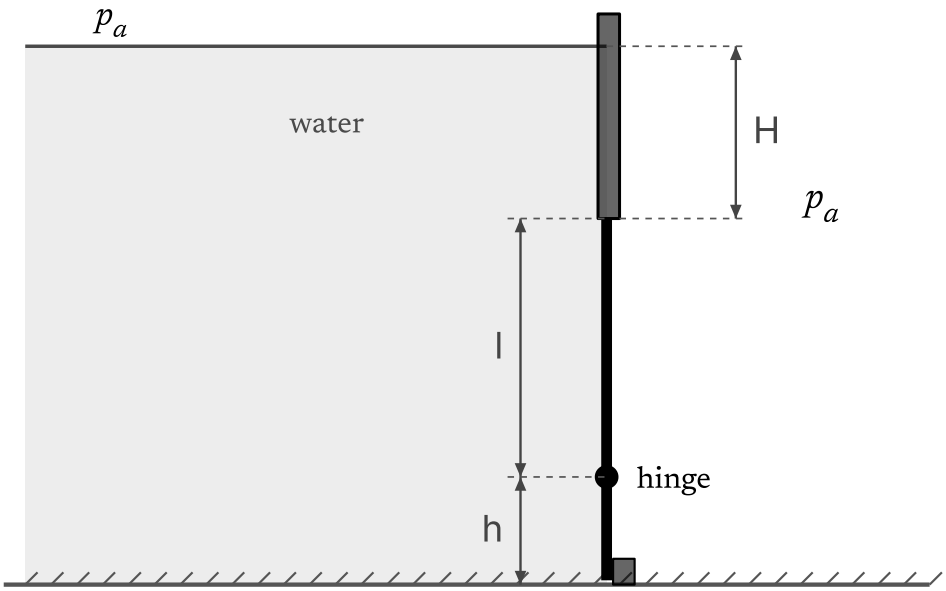
Original statement · fluid_statics_9.pdf

Plane surface | Moment about a hinge | Centre of pressure | Opening condition

OFFICIAL PROBLEM STATEMENT

Calculate the height H for which the hinged gate shown in the figure will open, if the hinge is at a distance l from the upper wall and at a known distance h from the floor ($h < l$).

Translated from the Spanish original. The upper wall is the fixed wall above the gate, so l is measured from the top edge of the gate.



Statement figure (redrawn from fluid_statics_9.pdf, p. 1): gate hinged at a distance l below its top edge and $h < l$ above the floor; the water surface is H above the top edge; air at p_a on both sides; a stop at the floor on the dry side

Source: fluid_statics_9.pdf (page 1, sheet of 10-09-04). Notes.pdf, Chapter 5: Eqs. 5.10, 5.11 and 5.13.

Worked solution

Hypotheses, Data and Unknowns

- Water at rest, incompressible, density ρ; air at p_a on the free surface and on the dry side, so only gauge pressures count.
- The gate is vertical, of height $l + h$ and width b, hinged without friction at a distance l below its top edge and h above the floor; its weight passes through the hinge and has no moment.
- A stop at the floor, on the dry side, keeps the lower edge from moving out: it can push the gate only towards the water.
- The free surface is at the height H above the top edge of the gate. Data: l, h, ρ, g, b. Unknown: the height H at which the gate opens.

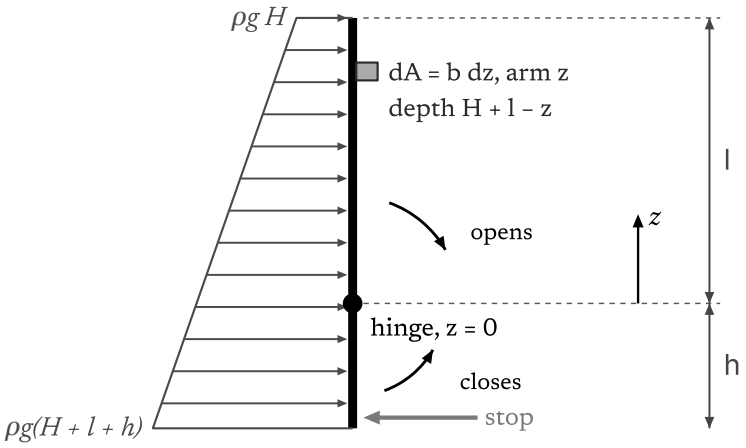
Symbol	Meaning	SI unit
H	water above the top edge of the gate	m
l, h	distances from the hinge to the top edge and to the floor	m
b	width of the gate	m

Coordinates and the Surface Element

Take z vertical, pointing up, with $z = 0$ at the hinge; the gate covers $-h \le z \le l$, and the free surface is at $z = H + l$. The depth of a point is $H + l - z$, so by (5.10)

$$p(z) - p_a = \rho g (H + l - z)$$

The strip between z and $z + dz$ has the area $b\,dz$ and the horizontal force $(p - p_a)\,b\,dz$ towards the dry side; its arm about the hinge is z. Strips above the hinge ($z > 0$) turn the gate with the top edge out (opening); strips below it ($z < 0$) turn it with the lower edge out, against the stop.



Free-body diagram of the gate (width b): the gauge pressure grows from $\rho g H$ at the top edge to $\rho g(H + l + h)$ at the floor; with z measured up from the hinge, the strip $b\,dz$ has depth $H + l - z$ and arm z. The part above the hinge turns the gate open; the part below presses it against the stop

Moment about the Hinge

Counting the opening sense as positive, the signed arm z takes care of both parts at once:

$$M = \int_{-h}^{l} \rho g(H + l - z)\, z\, b\, dz = \rho g b \left[(H + l)\frac{z^2}{2} - \frac{z^3}{3} \right]_{-h}^{l}$$

Barrow's rule, with $(-h)^2 = h^2$ and $(-h)^3 = -h^3$:

$$M = \rho g b \left[(H + l)\frac{l^2 - h^2}{2} - \frac{l^3 + h^3}{3} \right]$$

Opening Condition

While $M < 0$ the water presses the lower edge against the stop and the gate stays closed. It opens when M reaches zero and becomes positive; M grows with H because $l^2 - h^2 > 0$ ($h < l$). Setting $M = 0$:

$$(H + l)\frac{l^2 - h^2}{2} = \frac{l^3 + h^3}{3} \quad \Longrightarrow \quad H + l = \frac{2}{3}\frac{l^3 + h^3}{l^2 - h^2}$$

With $l^3 + h^3 = (l + h)(l^2 - lh + h^2)$ and $l^2 - h^2 = (l + h)(l - h)$:

$$H = \frac{2(l^2 - lh + h^2)}{3(l - h)} - l = \frac{2l^2 - 2lh + 2h^2 - 3l^2 + 3lh}{3(l - h)} = \frac{(l + h)(2h - l)}{3(l - h)}$$

The gate opens for any water level above this value.

Interpretation with the centre of pressure. $M = 0$ means that the resultant passes through the hinge: the centre of pressure is at the hinge. As H grows, the centre of pressure rises towards the centroid of the gate (Section 4.1 of Topic 5), which is $(l - h)/2$ above the hinge; the gate opens when the centre of pressure crosses the hinge from below.

Answers and Checks

- $H + l = \frac{2}{3}\frac{l^3 + h^3}{l^2 - h^2}$, that is $H = \frac{(l + h)(2h - l)}{3(l - h)}$.
- **When is it positive?** Only for $l/2 < h < l$. If $h \leq l/2$ the formula gives $H \leq 0$: even with the free surface at the top edge the centre of pressure is already above the hinge (for a triangular distribution it sits at a third of the height from the bottom, $(l + h)/3$ above the floor, which is above the hinge when $h < (l + h)/3$, i.e. $h < l/2$), so the gate cannot be kept closed. At $h = l/2$ the centre of pressure of the triangle is exactly at the hinge, $H = 0$.
- **Limiting case** $h \to l$ (hinge at the centre): $H \to \infty$; the centre of pressure is always below the centroid, so a gate hinged at its centre never opens (compare with FS4).
- **Dimensions**: H is m (cubic over quadratic lengths).

- **Check against the official key**: fluid_statics_9.pdf gives $M = \rho g[(H+l)(l^2-h^2)/2 - (l^3+h^3)/3]$ and $H + l = \frac{2}{3}(l^3+h^3)/(l^2-h^2)$, the same results.